

Assignment 03: Part 1. Document a chart

1. Documented Box Plot Code

Box Plot

Set the plot margins to control the amount of space around the plotting area
and prevent the labels and boxplots from being cut off.

```
par(mar = c(5, 4.1, 2, 1.5))
```

Create boxplots showing the distribution of tooth length for the three
Vitamin C dose levels for the ascorbic acid (VC) supplement.
boxwex controls the width of each box, at shifts the boxes slightly to
the left, col = "white" displays the VC boxes in white, and ylim sets
the vertical range of the plot.

```
boxplot(len ~ dose, data = ToothGrowth,  
        boxwex = 0.25, at = 1:3 - 0.2,  
        subset = supp == "VC", col = "white",  
        xlab = "",  
        ylab = "tooth length", ylim = c(0, 35))
```

Add the label for the x-axis and identify that the three groups represent
different Vitamin C dose levels measured in milligrams.

```
mtext("Vitamin C dose (mg)", side = 1, line = 2.5, cex = 0.8)
```

Add boxplots for the orange juice (OJ) supplement to the existing plot.
add = TRUE places the OJ boxplots on the same graph, while at shifts them
slightly to the right of the VC boxes so the two supplements can be compared
at each Vitamin C dose.

```
boxplot(len ~ dose, data = ToothGrowth, add = TRUE,  
        boxwex = 0.25, at = 1:3 + 0.2,  
        subset = supp == "OJ")
```

Add a legend to identify which boxplots represent ascorbic acid and
which represent orange juice, using the corresponding fill colors.

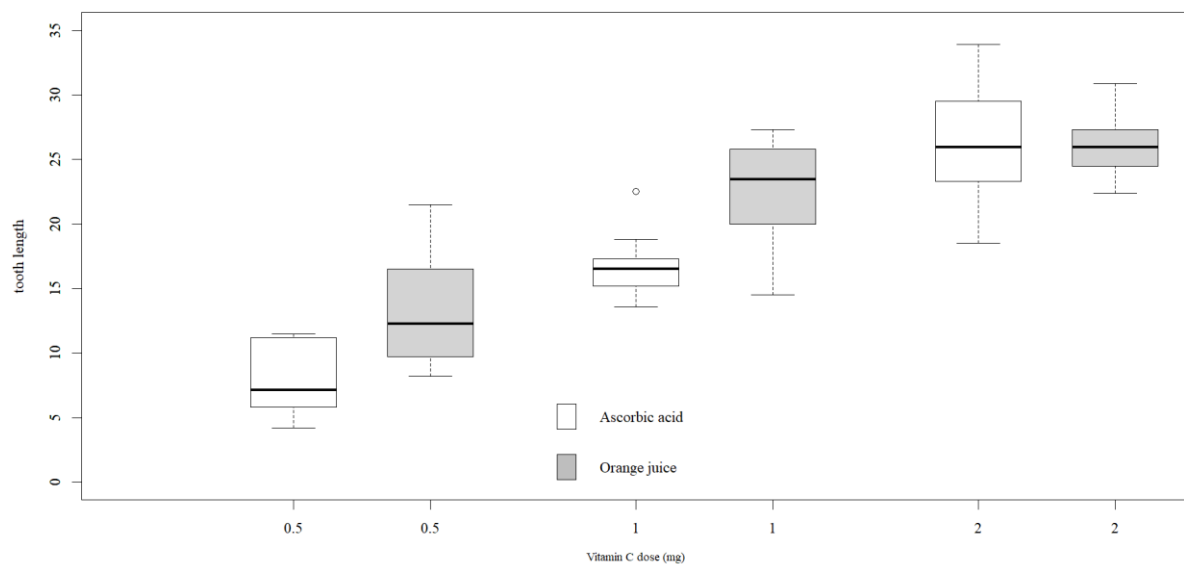
```
legend(1.5, 9, c("Ascorbic acid", "Orange juice"),  
      fill = c("white", "gray"),  
      bty = "n")
```

Restore the default plot margin settings after the box plot is completed.

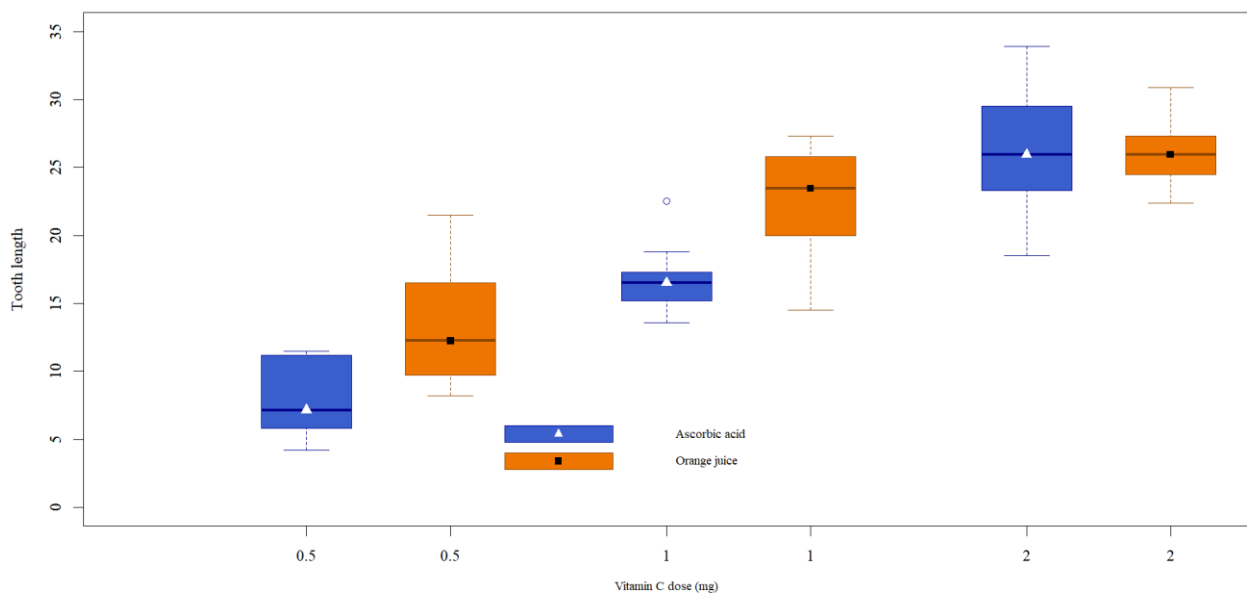
```
par(mar = c(5.1, 4.1, 4.1, 2.1))
```

2. Box Plot Description

I chose the box plot from the six charts. The figure compares tooth length across three Vitamin C dose levels for two types of supplements: ascorbic acid and orange juice. The first `boxplot()` call creates the boxplots for ascorbic acid, while the second `boxplot()` call adds the orange juice boxplots to the same figure. The `at` argument shifts the two supplement groups slightly to the left and right so that they can be compared at each dose level. The `boxwex` argument controls the width of the boxes. The `par()` function sets the margins of the plotting area, `mtext()` adds the x-axis label, and `legend()` identifies the two supplement types. The `ylim` argument sets the y-axis range from 0 to 35.



Assignment 03: Part 3. Finetune without leaving base R



The figure presents a fine-tuned boxplot of tooth length for different doses of Vitamin C and two types of supplements: ascorbic acid and orange juice. A serif font was used for the labels and text. Non-default colors, royal blue and dark orange, were used to distinguish the two supplement groups. A custom white triangle and black square were added to the legend to represent the plotting characters for the two groups. The plotting margins were adjusted using `par(mar = ...)` to ensure that the axis labels and other elements were clearly visible. All modifications were completed using base R graphics only.

CODE:

```
# Box Plot - Fine-tuned

# Set the margins: bottom, left, top, right
par(mar = c(5, 4.1, 2, 1.5),
    family = "serif")

# Create boxplots for Vitamin C supplement "VC"
boxplot(len ~ dose, data = ToothGrowth,
        boxwex = 0.25, at = 1:3 - 0.2,
        subset = supp == "VC",
        col = "royalblue3",
        border = "darkblue",
        xlab = "",
        ylab = "Tooth length",
        ylim = c(0, 35))

# Add the x-axis label
mtext("Vitamin C dose (mg)", side = 1, line = 2.5, cex = 0.8)

# Add boxplots for supplement "OJ"
boxplot(len ~ dose, data = ToothGrowth,
        add = TRUE,
        boxwex = 0.25, at = 1:3 + 0.2,
        subset = supp == "OJ",
        col = "darkorange2",
        border = "darkorange4")
```

```
# Add custom plotting characters to show the two supplement groups
```

```
points(1:3 - 0.2,  
       tapply(ToothGrowth$len[ToothGrowth$supp == "VC"],  
              ToothGrowth$dose[ToothGrowth$supp == "VC"], median),  
       pch = 17, col = "white", cex = 1.2)
```

```
points(1:3 + 0.2,  
       tapply(ToothGrowth$len[ToothGrowth$supp == "OJ"],  
              ToothGrowth$dose[ToothGrowth$supp == "OJ"], median),  
       pch = 15, col = "black", cex = 1.0)
```

```
# Legend
```

```
rect(1.35, 4.8, 1.65, 6.0,  
     col = "royalblue3", border = "darkblue")  
points(1.50, 5.4, pch = 17, col = "white", cex = 1)
```

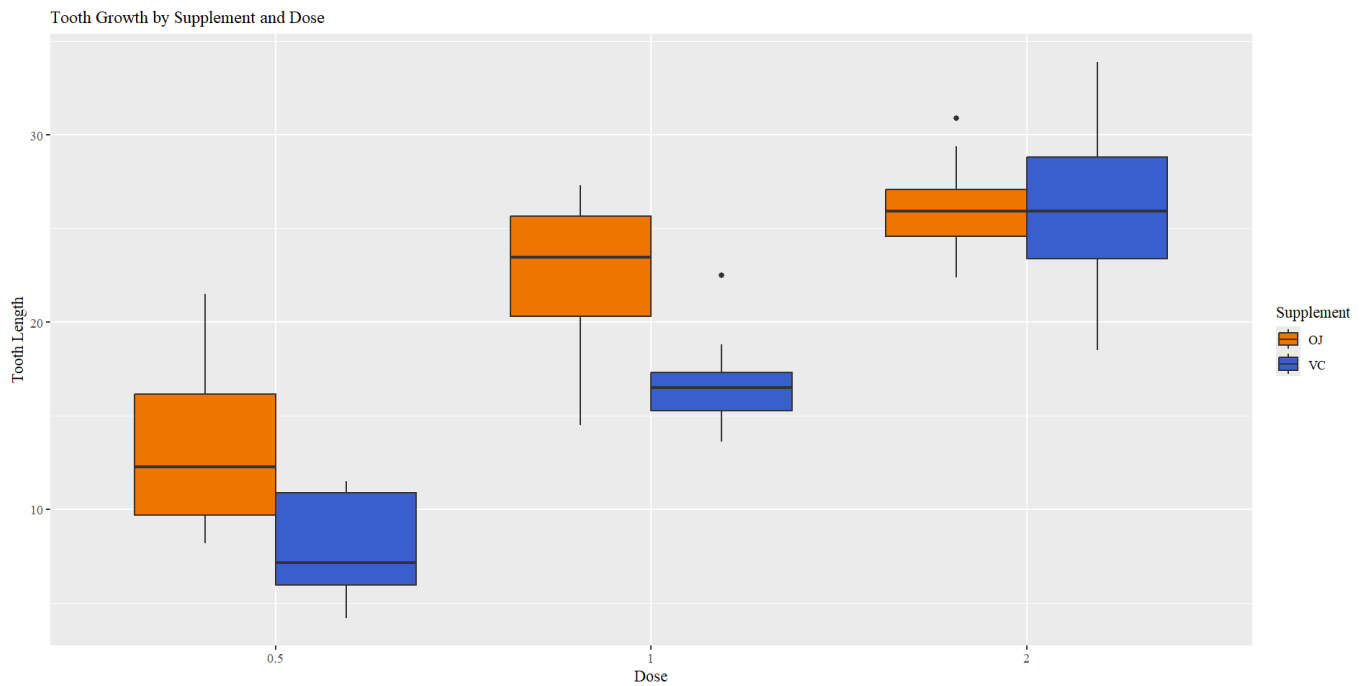
```
rect(1.35, 2.8, 1.65, 4.0,  
     col = "darkorange2", border = "darkorange4")  
points(1.50, 3.4, pch = 15, col = "black", cex = 1)
```

```
text(1.80, 5.4, "Ascorbic acid", pos = 4, cex = 0.8)  
text(1.80, 3.4, "Orange juice", pos = 4, cex = 0.8)
```

```
# Reset graphical parameters
```

```
par(mar = c(5.1, 4.1, 4.1, 2.1),  
    family = "sans")
```

Assignment 03: Part 4. Now do it in ggplot2



CODE

```
library(ggplot2)

# Create the boxplot using ggplot2
ggplot(ToothGrowth, aes(x = factor(dose), y = len, fill = supp)) +
  geom_boxplot(position = position_dodge(width = 0.75)) +

# Non-default colors
scale_fill_manual(values = c("VC" = "royalblue3",
                             "OJ" = "darkorange2")) +

# Labels
labs(
  title = "Tooth Growth by Supplement and Dose",
  x = "Dose",
  y = "Tooth Length",
  fill = "Supplement"
) +

# Serif font and adjusted margins
```

```
theme(  
  text = element_text(family = "serif"),  
  plot.title = element_text(size = 12),  
  plot.margin = margin(10, 15, 10, 10)  
)
```